

# HETVI CHIRAG SHETH

Machine Learning | Computer Vision | Agentic AI | Big Data | Third-Year B.Tech CSE

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## ABOUT ME

I'm a third-year Computer Science Engineering student at Navrachana University building AI systems end to end — backend, models, and UI. My flood-risk work spans two systems: RainWise, a Big Data pipeline (Spark, HDFS, MongoDB) forecasting flood risk with XGBoost and a Temporal Fusion Transformer, and a separate computer-vision system combining YOLOv8 and MobileNetV3 segmentation that reached 98.89% accuracy on hazard detection. I've also built Saarthi, a personal AI assistant with long-term vector memory; SecureInsight AI, an offline-only analytics platform with a local PyTorch anomaly detector; and an agentic CSV analyst built on LangGraph. I co-authored a research paper on smart waste management (GreenSort) presented at SustainX 2026 International Conference, and I've completed an IT internship at Lupin Limited. I like picking up whatever stack a project actually needs and shipping something that runs, not just a notebook.

## PROJECTS

**RainWise – Smart Rainfall Prediction System** : *Apache Spark, HDFS, MongoDB, XGBoost, TabTransformer, Temporal Fusion Transformer, Streamlit*

Link: <https://github.com/shethhetvi/rainwise>

- Built a Big Data flood-risk platform for Gujarat that ingests real-time rainfall, weather and river-gauge data and processes it through an Apache Spark + HDFS pipeline.
- Combined machine learning (XGBoost, logistic regression) with deep learning (TabTransformer, Temporal Fusion Transformer) for multi-day flood-risk forecasting.
- Stored enriched city-level summaries in MongoDB and served predictions through an interactive Streamlit dashboard.

**Flood Monitoring AI System** : *YOLOv8, LRASPP MobileNetV3, PyTorch, FastAPI, Streamlit, Docker (98.89% accuracy)*

Link: <https://github.com/shethhetvi/flood-monitoring-ai>

- Built a real-time computer-vision system for flood and urban-hazard monitoring, combining YOLOv8 object detection with LRASPP MobileNetV3 semantic segmentation.
- Reached 98.89% accuracy identifying hazards — water, roads, buildings and vegetation — from camera, video and image input.
- Shipped as both a Streamlit dashboard and a FastAPI service, containerized with Docker.

**Saarthi AI – Personal AI Mentor & Goal Planner** : *FastAPI, Streamlit, Supabase (PostgreSQL + pgvector), Gemini 2.5 Flash, Docker*

Link: <https://github.com/shethhetvi/Personal-AI-Assistant>

- Built a personal AI assistant with long-term semantic memory, using pgvector similarity search on Supabase to recall context across sessions.
- Designed a multi-agent architecture (planner, memory and recommendation agents) with a weighted scoring model to surface the single best next task.
- Added AI-generated goal roadmaps and interactive analytics, with JWT-based auth and a Dockerized deployment.

**SecureInsight AI** : *FastAPI, PyTorch, Next.js, TypeScript, AES-256*

Link: <https://github.com/shethhetvi/SecureInsightAI>

- Built a privacy-first analytics platform designed to run entirely offline, so client data never leaves the server.
- Implemented AES-256 encryption at rest and two-step OTP authentication, and trained a local PyTorch autoencoder to flag anomalies in uploaded datasets without any external API calls.
- Added a natural-language Q&A layer that translates plain-English questions into Pandas operations for secure data exploration.

**AI CSV Analyst** : *Streamlit, LangGraph, Gemini/Ollama, Pandas, Matplotlib*

Link: <https://github.com/shethhetvi/AI-CSV-Analyst>

- Built an agentic data-analysis tool that lets users upload a CSV and ask questions about it in plain English.
- Used LangGraph to orchestrate a routing agent that decides which tool — analysis, charting, or statistics — to call for a given question.
- Auto-generates Matplotlib visualizations and flags data-quality issues such as missing values and duplicates.

**Neural Transliterator – Hinglish to Devanagari** : *RNN, LSTM, BiLSTM, GRU, Attention, Transformer, CNN-LSTM, Flask*

Link: <https://github.com/shethhetvi/Transliterator>

- Built and benchmarked 8 transliteration architectures from scratch — rule-based, RNN, LSTM, BiLSTM, GRU, attention-based LSTM, a from-scratch Transformer, and a CNN-LSTM hybrid.
- Trained and evaluated all 8 models on the Dakshina Hindi transliteration dataset, comparing WER, CER and accuracy across architectures as a 4-person team.
- Packaged the best-performing model into a Flask web app for live Devanagari transliteration.

**GreenSort – Smart Waste Management System** : *Team project with Sneha Patel and Parisha Parmar*

- Worked with Sneha Patel and Parisha Parmar to build GreenSort, a smart waste management system.
- Co-authored and presented a research paper on the project at SustainX 2026 International Conference, CVM University.

**TransitOps – Transport Operations Management Platform** : *React, Node.js, PostgreSQL, Prisma, REST APIs — Odoo Hackathon 2026*

- Team member on TransitOps, a centralized platform for managing transport operations and lifecycle workflows, built at the Odoo Hackathon 2026.
- Contributed across frontend, backend and database layers to bring operational workflows into one unified system under hackathon time constraints.

Other projects: MedVision AI – Intelligent Medical Image Diagnosis Platform · PortfolioStrength – AI portfolio evaluator SaaS with an AI-recruiter chatbot ([github.com/shethhetvi/Data-Analyst-Portfolio-Website](https://github.com/shethhetvi/Data-Analyst-Portfolio-Website)) · BakeMyDay – Online Cake Ordering System (Java, JDBC, JSP) · Adaptive Microlearning Platform (TetraTHON 2026 EdTech Hackathon).

## EXPERIENCE

**IT Intern** — Lupin Limited (Pharmaceutical)

09 Sep 2022 – 22 Sep 2022

- Completed an internship in the IT department as part of the engineering curriculum, gaining exposure to IT operations and technical support.
- Recognized for being sincere, hardworking and regular throughout the internship.

## EDUCATION

**B.Tech – Computer Science Engineering**

Navrachana University, Vadodara

2023 – 2027 · CGPA: 8.55 / 10

**Diploma – Computer Engineering**

Sigma Institute of Engineering, Vadodara (GTU)

2021 – 2023 · CGPA: 9.86 / 10

Diploma Distinction, zero backlogs across all 8 semesters

## SKILLS

**AI / Machine Learning:** PyTorch, XGBoost, TabTransformer, Temporal Fusion Transformer, YOLOv8, LRASPP MobileNetV3, RNN/LSTM/BiLSTM/GRU/Transformer architectures, Deep Learning, Computer Vision, NLP

**Generative & Agentic AI:** LangGraph, Google Gemini API, Vector Search (pgvector), Retrieval-Augmented workflows

**Big Data & Cloud:** Apache Spark, HDFS, MongoDB, Supabase (PostgreSQL), AWS

**Backend / Web:** FastAPI, Flask, Streamlit, Next.js, TypeScript

**Data Analytics & Visualization:** Power BI, Tableau, Pandas, Matplotlib, Data Analysis, Data Visualization, Dashboards

**Security:** AES-256 encryption, JWT authentication, OTP-based 2FA

**Programming Languages:** Python, Java, JavaScript

**IoT:** Internet of Things (IoT)

## CERTIFICATIONS

Data Analysis: Deloitte – Data Analytics Job Simulation (Forage) · IBM – Python for Data Science · IBM Cognitive Class – Data Analysis with Python

IoT: Cisco Networking Academy – Introduction to Internet of Things and Digital Transformation

## ACHIEVEMENTS & LEADERSHIP

- Co-authored and presented a research paper on GreenSort (smart waste management) at SustainX 2026 International Conference, CVM University.
- Achieved 98.89% accuracy with the Flood Monitoring AI System (multi-hazard flood detection).
- Diploma Distinction with CGPA 9.86 and zero backlogs across all 8 semesters.
- AI Supplier Ranking Agent appreciated by the CEO of Mesh Works during a hackathon.
- Participant, TetraTHON 2026 EdTech Hackathon — adaptive microlearning platform.

## LANGUAGES

Gujarati · Hindi · English